

Siddhartha Prasad

siddhartha.a.prasad@gmail.com | siddharthaprasad.com

I am a researcher focused on helping people write programs that behave as they intend. My research interests are informed by my time as an engineer: I have written code that doesn't do what I want it to, and I want to spare everyone else the indignity.

EDUCATION

Brown University Providence, RI Jan 2022 – present
Ph.D. in Computer Science

Tufts University Medford, MA May 2016
B.S. in Computer Science and Mathematics
Summa cum Laude; High Thesis Honors

EMPLOYMENT

Apple Seattle, WA
Research Intern · Human-Centered Machine Learning May – Sept 2023

Explored how AI and formal-methods techniques can support early childhood education.

Microsoft Redmond, WA
Software Engineer II · Applied AI April 2018 – Sept 2021

- Developer on the Azure Cognitive Services team, which lets developers embed AI into apps, bots, and websites.
- Built a containerized framework that allowed AI models to be run portably across operating systems and architectures.

Software Engineer · Developer Ecosystem Platform Aug 2016 – March 2018

- Developed APIs and features for XAML, a UI language used to build Universal Windows apps.
- Designed XAML APIs for input modalities (keyboard, gamepad, ink), accessibility, and cross-platform language support.

Software Engineering Intern · Developer Ecosystem May – Aug 2015

INRIA Saclay, France
Research Intern · Parsifal June – July 2014

Research on correctness certificates for first-order term rewriting.

PUBLICATIONS

Diagramming Program Values by Spatial Refinement
Siddhartha Prasad, Michael Tu, Karan Kashyap, Tim Nelson, Shriram Krishnamurthi
[To appear in] PLDI 2026

hS: Speculative Script Reordering at Subprocess Granularity
Georgios Liargkovas, Di Jin, Tianyu (Ezri) Zhu, Dan Liu, A. Bolun Thompson, Anirudh Narsipur, Seong-Heon Jung,
Siddhartha Prasad, Diomidis Spinellis, Michael Greenberg, Konstantinos Kallas, Nikos Vasilakis
[To appear in] OSDI 2026

Meaningful Human-in-the-Loop Checking of GenAI Synthesis for Restricted Languages*Siddhartha Prasad*, Skyler Austen, Kathi Fisler, Shriram Krishnamurthi**[To appear in]** *ECOOP 2026***A Misconception-Driven Adaptive Tutor for Linear Temporal Logic***Siddhartha Prasad*, Ben Greenman, Tim Nelson, Shriram KrishnamurthiCAV 2025 **[Distinguished Paper]****Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design***Siddhartha Prasad*, Ben Greenman, Tim Nelson, Shriram Krishnamurthi*ECOOP 2025* **[Distinguished Paper; Distinguished Artifact]****Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic**Ben Greenman, *Siddhartha Prasad*, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, Milda Zizyte*FM 2024***ContextQ: Generated Questions to Support Meaningful Parent-Child Dialogue While Co-Reading**Griffin Dietz Smith, *Siddhartha Prasad*, Matt J Davidson, Leah Findlater, R Benjamin Shapiro*IDC 2024***Forge: A Tool and Language for Teaching Formal Methods**Tim Nelson, Ben Greenman, *Siddhartha Prasad*, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, Shriram Krishnamurthi*OOPSLA 2024***Conceptual Mutation Testing for Student Programming Misconceptions***Siddhartha Prasad*, Ben Greenman, Tim Nelson, Shriram Krishnamurthi*Programming 2024***Generating Programs Trivially: Student Use of Large Language Models***Siddhartha Prasad*, Ben Greenman, Tim Nelson, Shriram Krishnamurthi*CompEd 2023***Making Hay from Wheats: A Classsourcing Method to Identify Misconceptions***Siddhartha Prasad*, Ben Greenman, Tim Nelson, John Wrenn, Shriram Krishnamurthi*Koli Calling 2022***Large-Scale Intelligent Microservices**Mark Hamilton, Nick Gonsalves, Christina Lee, Anand Raman, Brendan Walsh, *Siddhartha Prasad*, Dalitso Banda, Lucy Zhang, Lei Zhang, William T Freeman*IEEE BigData 2020***TEACHING**

Teaching Assistant

2022–2026 CSCI 1710 — Logic for Systems

Brown University

2025 CSCI 1730 — Programming Languages

Brown University

2015–2016 CS 105 — Programming Languages

Tufts University

2014 CS 160 — Algorithms

Tufts University

SERVICE

2026 Artifact Reviewer, *The Art, Science, and Engineering of Programming*

AFFILIATIONS & CERTIFICATIONS

Sheridan Teaching Seminar Certificate
Tau Beta Pi, The Engineering Honor Society

Brown University
Tufts University